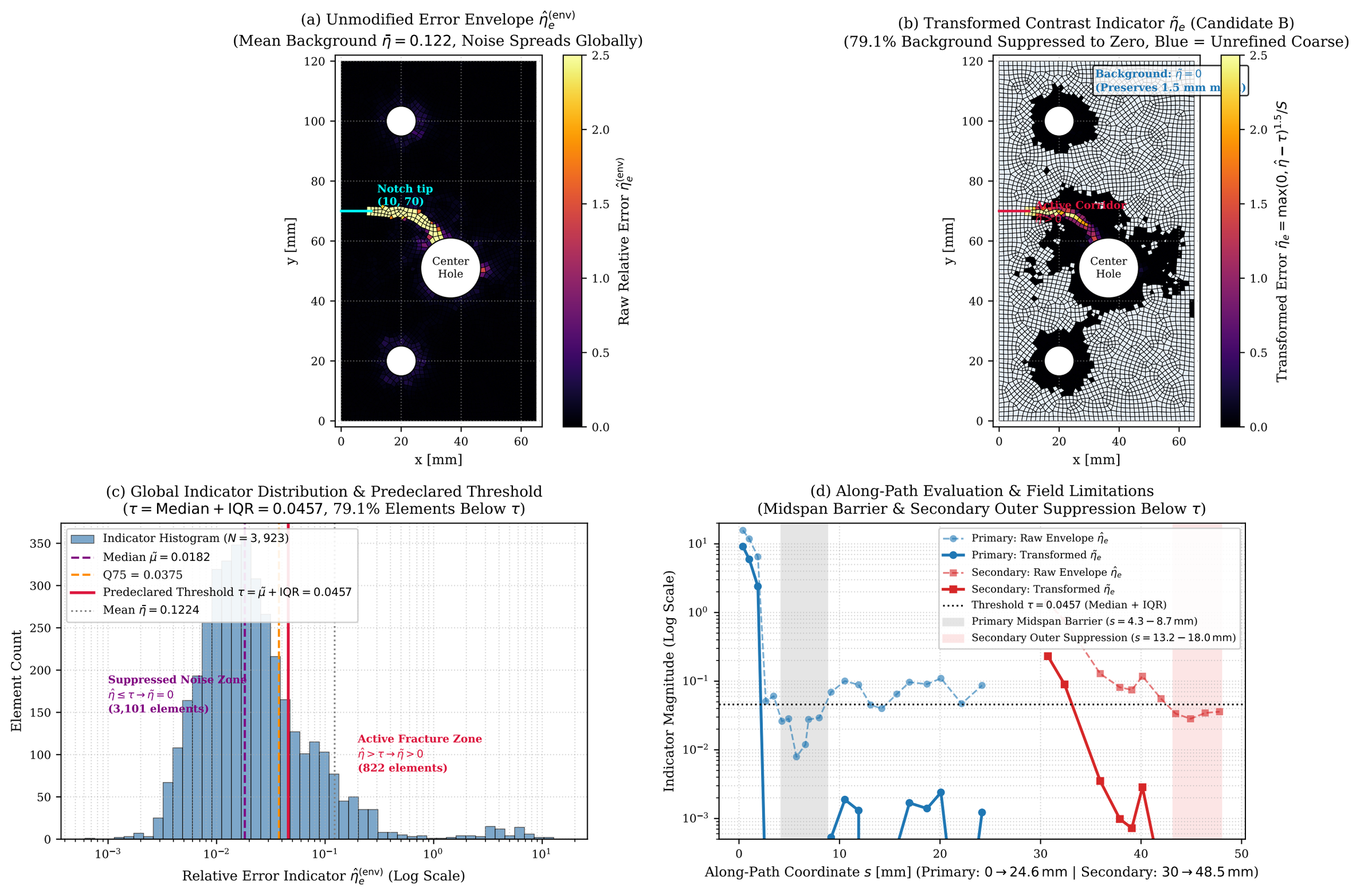

(a) Unmodified Error Envelope $\hat{\eta}_e^{(\mathrm{env})}$
(Mean Background $\bar{\eta} = 0.122$, Noise Spreads Globally)
Notch tip
(10, 70)
Center
Hole
x [mm]
y [mm]
Raw Relative Error $\hat{\eta}_e^{(\mathrm{env})}$
(b) Transformed Contrast Indicator $\tilde{\eta}_e$ (Candidate B)
(79.1% Background Suppressed to Zero, Blue = Unrefined Coarse)
Background: $\tilde{\eta} = 0$
Active Corridor
$\tilde{\eta} > 0$
Center
Hole
x [mm]
y [mm]
Transformed Error $\tilde{\eta}_e = \max(0, \hat{\eta} - \tau)^{1.5}/S$
(c) Global Indicator Distribution & Predeclared Threshold
(τ = Median + IQR = 0.0457, 79.1% Elements Below τ)
Indicator Histogram ($N = 3,923$)
Median $\tilde{\mu} = 0.0182$
Q75 = 0.0375
Predeclared Threshold $\tau = \tilde{\mu} + \mathrm{IQR} = 0.0457$
Mean $\bar{\eta} = 0.1224$
Suppressed Noise Zone
$\hat{\eta} \le \tau \to \tilde{\eta} = 0$
(3,101 elements)
Active Fracture Zone
$\hat{\eta} > \tau \to \tilde{\eta} > 0$
(822 elements)
Element Count
Relative Error Indicator $\hat{\eta}_e^{(\mathrm{env})}$ (Log Scale)
(d) Along-Path Evaluation & Field Limitations
(Midspan Barrier & Secondary Outer Suppression Below τ)
Primary: Raw Envelope $\hat{\eta}_e$
Primary: Transformed $\tilde{\eta}_e$
Secondary: Raw Envelope $\hat{\eta}_e$
Secondary: Transformed $\tilde{\eta}_e$
Threshold $\tau = 0.0457$ (Median + IQR)
Primary Midspan Barrier ($s = 4.3 - 8.7$ mm)
Secondary Outer Suppression ($s = 13.2 - 18.0$ mm)
Indicator Magnitude (Log Scale)
Along-Path Coordinate s [mm] (Primary: $0 \to 24.6$ mm | Secondary: $30 \to 48.5$ mm)